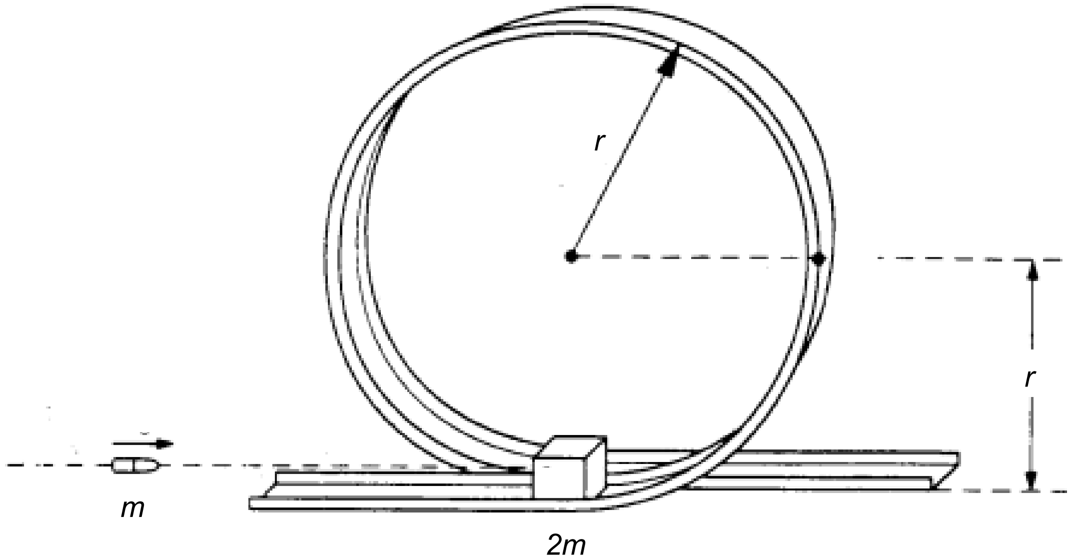
2	A block of mass $2m$ initially rests on a track at the bottom of the circular, vertical loop of radius r as shown in Fig. 2.1. A bullet of mass m strikes the block horizontally and remains embedded in the block as the block and bullet circle the loop. Assume frictional force of the loop is negligible.  **Fig. 2.1**			
	(a)	**(i)**	Show that the minimum speed of the block at the top of the loop such that it just completes the vertical circular motion without falling off the loop is given by $\sqrt{gr}$ where g is the acceleration due to gravity.	[3]
			Solution: At the **Top** of the loop, the **vector sum** of the **Weight of the block (with bullet embedded)** ***[3mg]*** and the **Normal contacf force *[N]* exerted by the loop on the block** <u>**provide for**</u> **the Centripetal force required** $\left[\frac{(3m)v^2}{r}\right]$.	**1**
			Thus, $3mg + N = \frac{(3m)v^2}{r}$ **where *v* is the speed at the top**	**1**
			$N = \frac{(3m)v^2}{r} - 3mg$	
			For the block **to JUST complete** the vertical circular motion without falling off the loop **at the Top, *N* ≥ *0*.** $\Longrightarrow \frac{(3m)v^2}{r} - 3mg \geq 0$ $\Longrightarrow v \geq \sqrt{rg}$ Minimum velocity at the top of the loop, $v_{min,top} = \sqrt{rg}$	**1**

		(ii)	Derive an expression for the minimum speed of the bullet in order for the block to just complete the vertical circular motion without falling off the loop. Explain your working.	[4]
			Solution: Since the track is frictionless, the loss in the kinetic energy of the block and bullet is converted into their gain in gravitational potential energy. Therefore, $\frac{1}{2}(3m)\,v_{min,top}{}^2 - \frac{1}{2}(3m)v_{bottom}{}^2 = (3m)g(2r)$ $\frac{1}{2}(3m)\left(\sqrt{rg}\right)^2 - \frac{1}{2}(3m)v_{bottom}{}^2 = (3m)g(2r)$ $v_{bottom} = \sqrt{5rg}$ where v_{bottom} is the speed at the **bottom** of the loop Applying principle of conservation of linear momentum, taking vectors to the right as positive, Total initial momentum of bullet = Total momentum of block with embedded bullet JUST AFTER the collision $mv_{bullet} = 3m\sqrt{5rg}$ $v_{bullet} = 3\sqrt{5rg}$	1 1 1 1

3	(a)	A ball is held between two fixed points A and B by means of two stretched springs as shown in